# Do Managers Learn from Market Reactions? A Replication of Jayaraman and Wu (2020)

*Should I Stay or Should I Grow? Using Voluntary Disclosure to Elicit Market Feedback*, Review of Financial Studies 33(8): 3854–3888

## 1 Overview of Results

We replicate Tables 1–3 of the paper with WRDS data. The findings fall into three layers. First, the paper's association replicates: the better the market reaction to a capex forecast, the more managers subsequently revise capex upward. Second, the key evidence that managers learn from informed traders—that the effect appears only for forecasts that attract informed trading—does not replicate. Third, this difference cannot be attributed to the spread measure, data backfilling or sample filters. The paper's pattern appears only partly, in forecasts from 2009 onward or under the variable definitions closest to the paper, and the difference between groups is never significant at the 5% level.

The paper argues in three steps: it establishes an association, uses two cross-sectional tests to rule out omitted variables and reverse causality, and then draws a causal interpretation. Table 1 summarizes the replication of each step, and Figure 1 shows how they connect to the conclusions.

**Table 1. Overview of replication results**

| Evidence | Paper | This replication | Verdict |
|---|---|---|---|
| ① Market reaction predicts capex adjustment (Table 2) | CAR significant and positive in all 8 models | Significant in all 8 models, within 1 SE of the paper | Holds |
| ② Only when spreads rise, i.e. informed trading (Table 3, cols 1–2) | Lower −0.295; higher 1.206**; difference p = 0.019 | Lower 0.674**; higher 0.880**; difference p = 0.672 | Not replicated |
| ③ Driven by positive reactions (Table 3, cols 3–4) | CAR_POS difference p = 0.099 | Same sign; difference p = 0.232 | Weak |

Notes: ** p < 0.05. CAR is standardized: coefficients are CAPEX_ADJ percentage points per one standard deviation of CAR.

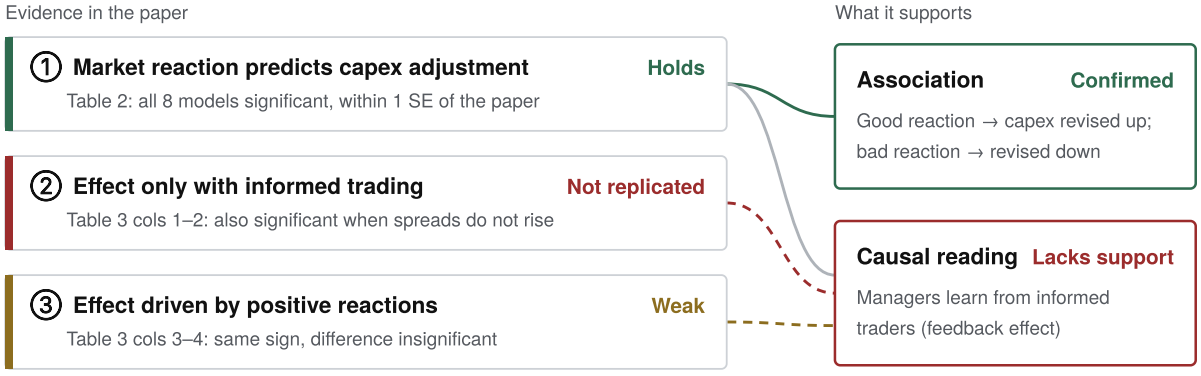



**Figure 1. The paper's chain of evidence and the replication**

Notes: With ② not replicated and ③ weak, the causal reading loses its main support; ① alone supports only an association. Appendix A shows how Tables 4–6 are affected.

## 2 Scope and Method

We replicate Tables 1–3, which test the paper's first hypothesis. All benchmark numbers are taken from the published version and checked item by item: coefficients, standard errors, sample sizes, $R^2$ and p-values. Table 2 lists the data and implementation and Table 3 the main variables.

**Table 2. Data and implementation**

| Item | Details |
|---|---|
| Data | I/B/E/S Guidance (capex, sales and earnings-type guidance) and actual EPS announcement dates; Compustat annual and quarterly; CRSP daily returns and quotes; CRSP–Compustat link table. Downloaded from WRDS in November 2026 |
| Estimation | High-dimensional fixed-effects regressions with clustered standard errors in Python, following the Stata `reghdfe` conventions; verified against full dummy-variable regressions to 1e−8 |
| Unstated details | 37 in total, each a switch (D1–D37); the 23 relevant to Tables 1–3 are listed in Appendix F |

**Table 3. Main variables**

| Variable | Definition |
|---|---|
| CAPEX_ADJ | Deviation of actual from forecast capex (%); zero when actual capex falls inside a range forecast |
| CAR | Cumulative abnormal return over days [−2, 2] around the forecast (firm minus S&P 500), standardized in regressions |
| CAR_POS / CAR_NEG | Positive and negative parts of CAR |
| Higher / lower spreads | Whether the relative bid–ask spread in [−2, 2] exceeds that in [−10, −3]; rising spreads mark informed trading |
| EA_DUM | Indicator for capex forecasts issued on the day of an earnings announcement |

Three principles guide the implementation. First, choices are calibrated on sample characteristics only, never on regression results. Second, all diagnostic tests and decision rules were registered before they were run. Third, we do not search over specifications for significance, and every test is reported whatever its outcome.

## 3 Sample Construction

Our final sample has 18,866 forecasts from 2,180 firms, against 17,577 forecasts from 1,790 firms in the paper: 7.3% more forecasts and 21.8% more firms. Table 4 compares the filtering steps.

**Table 4. Sample filtering steps**

| Step | Paper | Ours | Difference |
|---|---|---|---|
| Matched to Compustat actual capex | 40,785 | 41,452 | +1.6% |
| With a 5-day CRSP market reaction | 36,900 | 41,443 | +12.3% |
| Excluding forecasts concurrent with earnings forecasts | 19,427 | 20,344 | +4.7% |
| Excluding forecasts made in 2016 (final sample) | 17,577 | 18,866 | +7.3% |

Notes: We require an I/B/E/S–CRSP link at the first step, so the 3,885 forecasts the paper drops at its second step never enter our first step. The counts are comparable after the second step.

The gap is concentrated early in the sample (Figure 2): we have 939 more forecasts in 2003–2009 and 350 more in 2010–2015. The 390 extra firms are also mostly early first-time forecasters.

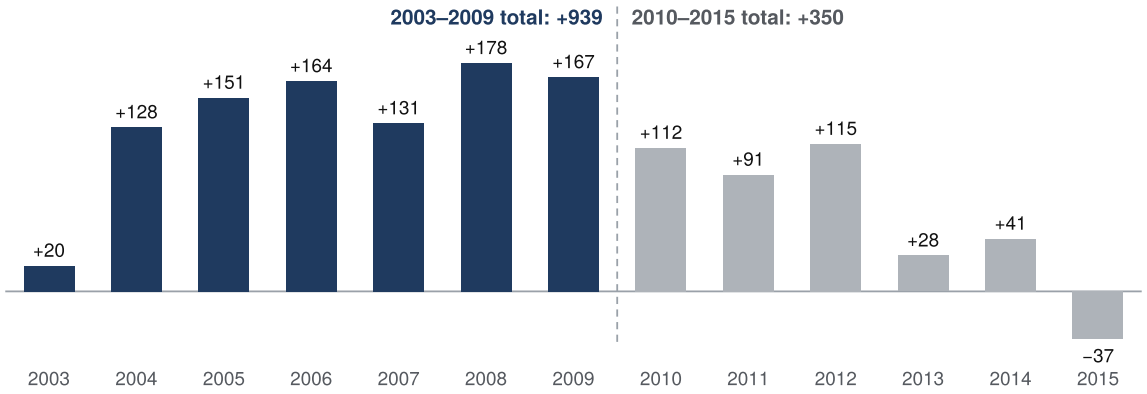



**Figure 2. Difference in forecasts by year (ours minus paper)**

Sources: the paper's Table 1 Panel B; `annual_waterfall.py`.

The annual waterfall (Appendix C) locates the gap by year and step. Our retention rates at the first four steps are stable across years: about 0.88–0.93 when keeping one record per announcement, 0.93–0.97 at the Compustat match and close to 1 at the CRSP match. Only the step that removes forecasts concurrent with earnings forecasts varies with the year, rising from 0.29 in 2003 to 0.57 in 2009.

Figure 3 compares two ratios. From 2009 on, the paper's final count divided by our step-4 count (about 0.50) matches our step-5 retention (about 0.49–0.57), so the two pipelines then produce similar results. In 2003–2008 the paper keeps a much smaller share (0.15–0.42 against our 0.29–0.47). The ratio of the paper's final count to each of our steps rises in the same way from the raw records onward, while our first four steps treat all years alike. The early gap therefore does not arise at any of our steps but at a step in the paper without a counterpart in ours, one that removes more forecasts in early years. The likeliest candidate is the paper's CRSP match, where it drops 3,885 forecasts against our 9. The paper does not report this step by year, so this cannot be confirmed.

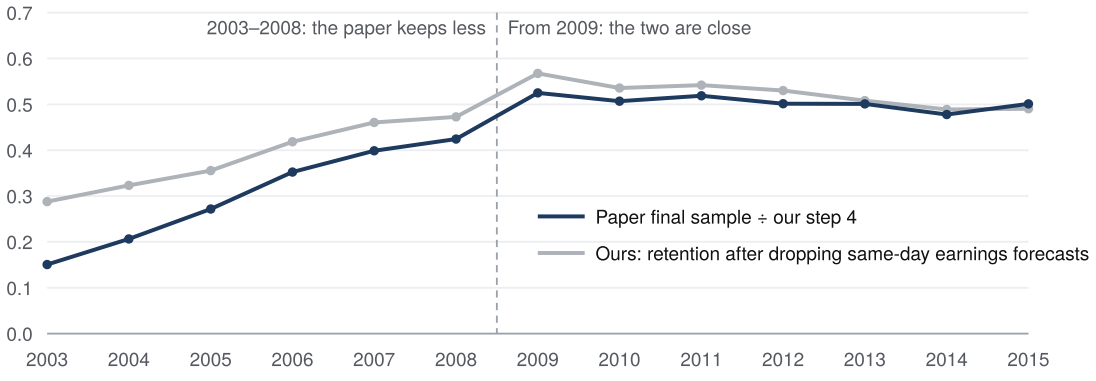



**Figure 3. Annual retention ratios: paper and replication**

Notes: Step 4 is "with a CRSP market reaction"; step 5 removes forecasts concurrent with earnings forecasts. Source: `annual_waterfall.py`.

## 4 Replication of Tables 1 and 2

The core result of Table 2 replicates (Table 5; full comparison in Appendix B). The CAR coefficient is significant and positive in all eight models and within one standard error of the paper's estimate in each; as in the paper, the three placebo variables are insignificant. All robustness checks reported in the text hold. Numbers quoted in the text also match: the median gap between sales and capex forecasts is 86 days, as in the paper, and the median market reaction to sales forecasts is 0.447% (paper: 0.457%).

**Table 5. Table 2: main coefficients (dependent variable: CAPEX_ADJ)**

| Model | Paper | Ours |
|---|---|---|
| (1) No controls, no fixed effects | 1.193*** | 1.193*** |
| (4) Firm and year-quarter fixed effects | 0.975*** | 1.017*** |
| (8) All controls | 0.849*** | 0.988*** |
| Placebos: SF_CAR, MKTRET, CAR_PRE | All insignificant | All insignificant |
| Footnote 20: industry × year / state × year fixed effects | 0.800 / 0.968 | 0.779 / 0.752 |

Notes: *** p < 0.01. Models (4)–(8) include firm and year-quarter fixed effects; standard errors in (5)–(8) are clustered by industry.

Some descriptive statistics, however, differ from the paper under every specification (Table 6); Section 7 examines why. Because our mean CAPEX_ADJ is lower, the same coefficient looks larger relative to the mean; what replicates is the effect measured in percentage points.

**Table 6. Descriptive statistics that differ from the paper (Table 1)**

| Variable | Paper | Ours |
|---|---|---|
| EA_DUM (share issued on earnings announcement days) | 0.559 | 0.662 |
| Mean CAR (%) | −0.148 | 0.318 |
| CAPEX_ADJ mean / median (%) | 7.421 / 0.000 | 3.945 / −0.326 |
| Point / range / others | 55.3% / 34.5% / 10.2% | 54.6% / 42.6% / 2.8% |

## 5 Replication of Table 3

The paper uses Table 3 to read the association as feedback. Rising spreads indicate that a forecast attracted informed trading; if the effect appears only for those forecasts, omitted variables and reverse causality struggle to explain the pattern. We find the effect both when spreads rise and when they do not, and the difference between the groups is insignificant (Table 7).

**Table 7. Table 3 (dependent variable: CAPEX_ADJ)**

| | Paper: lower | Paper: higher | Ours: lower | Ours: higher |
|---|---|---|---|---|
| CAR | −0.295 | 1.206** | 0.674** | 0.880** |
| | (0.372) | (0.497) | (0.327) | (0.350) |
| p-value of difference | | 0.019 | | 0.672 |
| CAR_POS | −0.203 | 0.893* | 0.122 | 0.676* |
| | (0.419) | (0.455) | (0.360) | (0.388) |
| CAR_NEG | −0.045 | 0.185 | 0.803* | 0.412 |
| | (0.176) | (0.129) | (0.455) | (0.394) |
| p-value of difference (CAR_POS) | | 0.099 | | 0.232 |
| N | 7,770 | 9,570 | 8,135 | 10,542 |

Notes: Standard errors clustered by industry in parentheses. Controls and fixed effects as in Table 2 model (8). * p < 0.1, ** p < 0.05. Full comparison in Appendix B.

For CAR, our result disagrees with the paper. The difference is 0.206 (standard error 0.485) with a 95% confidence interval of [−0.76, 1.18], which excludes the paper's 1.501 (p = 0.010). Allowing for the paper's own uncertainty as well, the p-value of the gap between the two estimates is 0.107; because the two samples overlap heavily, this p-value is conservative.

For CAR_POS, the sign agrees with the paper but power is low: the difference is 0.553 with a 95% interval of [−0.36, 1.47], which contains the paper's 1.096 (p = 0.24). Note also that the effect in our lower-spread group comes mainly from negative reactions (CAR_NEG 0.803, p = 0.08). The paper notes that a disciplining channel would predict stronger effects for negative reactions; this result is consistent with that channel but significant only at the 10% level.

## 6 Why Table 3 Differs

Before running any test, we registered three candidate explanations and what each predicts for every test. The decision rule requires at least two related tests to agree, since one significant result in seven tests could easily arise by chance. Table 8 reports the verdicts and Figure 4 the group differences in each test.

**Table 8. Pre-registered verdicts**

| Explanation | Rule | Result | Verdict |
|---|---|---|---|
| E1 Noisy spread measure: splitting closing quotes misclassifies forecasts | The continuous test (B) and the extreme-tercile test (C) both show a stronger effect | B: interaction 0.071, p = 0.79; C: difference 0.56, p = 0.44 | Rejected |
| E2 Different sample composition or data vintage | The 2009-onward subsample (D1) and the sample closest to the paper (E) both show the paper's pattern | D1 does (p = 0.058); E does not (difference −0.22) | Not met |
| E3 The paper's pattern is not robust in current data | Main interval excludes 1.501 and no other test shows the pattern | Main test meets it (p = 0.010); D1 is the exception | Largely met |

Notes: Full results in Appendix E.

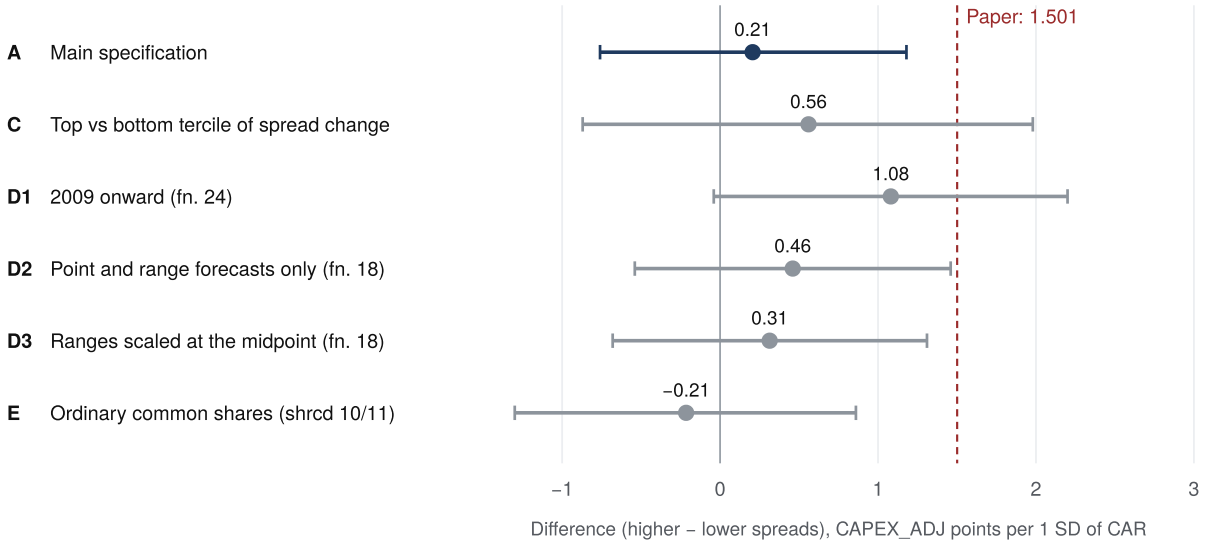



**Figure 4. Group differences and 95% confidence intervals by test**

Notes: Only the intervals for C and D1 contain the paper's 1.501; C's interval also contains zero, and D1's lower bound is close to zero. Test B estimates an interaction and is not shown. Source: `t3_diagnostics.py`.

Two further findings narrow the field. First, the gap does not come from I/B/E/S backfilling: of 326,000 guidance records, only 3 were activated after the paper's submission date of 21 July 2018, and the median lag between activation and announcement is 0–6 days in every year. Second, firm-level filters explain only part of the sample gap: keeping ordinary common shares (shrcd 10/11) removes 10.9% of forecasts and 14.9% of firms, closer to the paper's 6.8% and 17.9%, yet on that sample Table 3 looks even less like the paper.

## 7 Forecast-Level Definitions

A forecast-level difference means that the same I/B/E/S record is coded differently in the paper and in our replication, rather than that the two samples contain different firms. The evidence: whatever firm-level filter we apply—share code, exchange, listing age or link quality—EA_DUM stays at 0.662, the share of "others" at 2.8%, and the means of CAR and CAPEX_ADJ barely move. These statistics depend only on how each forecast is defined and computed. The paper leaves three forecast-level definitions unstated, and we test three or four alternatives for each (Table 9; all variants in Appendix D).

**Table 9. Tests of forecast-level definitions**

| Definition | Alternatives | Result |
|---|---|---|
| Forecast type (D3) | Codes 06, 08 ("high/low end of") as others; classify by I/B/E/S code only | Codes 06 and 08 cover only 42 forecasts (0.2%). All open-ended codes in current I/B/E/S total 2.8%, so no code-based rule yields the paper's 10.2% |
| EA_DUM (D10) | I/B/E/S actual EPS announcement date; same day or ±1 day | I/B/E/S dates almost coincide with Compustat report dates (0.665 vs 0.662); ±1 day raises EA_DUM to 0.90 |
| "Earnings forecast" (D9) | EPS only; any other guidance on the same day; earnings-type within ±1 day | Counting any other same-day guidance cuts firms to 1,908 (paper: 1,790), sets the CAPEX_ADJ median to 0.000 (as in the paper), raises the mean to 5.56 (paper: 7.42) and lowers EA_DUM to 0.633; but it removes 59.9% of forecasts against the paper's 47.4% |
| Mean CAR | Reweight by EA_DUM; use the CRSP value-weighted index | Mean CAR is nearly identical with and without an earnings announcement (0.313 vs 0.327); the value-weighted index gives 0.336. Neither reaches the paper's −0.148 |

The results reject the lead suggested in the previous draft: codes 06 and 08 are far too rare to explain the gap in the range and "others" shares. The paper's 10.2% of "others" cannot be reproduced from current I/B/E/S codes and may reflect an older coding or a manual classification. Nor can any alternative definition explain the gaps in EA_DUM and mean CAR. The only change that moves several statistics toward the paper is the definition of an earnings forecast: counting any other same-day guidance lowers the overall distance to the paper from 1.484 to 1.080. That definition removes too many forecasts, however, so the authors' choice probably lies between the two.

As pre-registered, we estimate Table 3 once on the combination with the smallest distance (Table 10).

**Table 10. Table 3 (CAR) under the definitions closest to the paper**

| | Lower | Higher | Difference | p | 95% CI |
|---|---|---|---|---|---|
| Paper | −0.295 (0.372) | 1.206** (0.497) | 1.501 | 0.019 | — |
| Our main specification | 0.674** (0.327) | 0.880** (0.350) | 0.206 (0.485) | 0.672 | [−0.76, 1.18] |
| Closest definitions | 0.340 (0.343) | 1.032** (0.445) | 0.692 (0.561) | 0.222 | [−0.43, 1.81] |

Notes: "Closest definitions" treats any other same-day guidance as an earnings forecast, classifies forecast types by I/B/E/S code, and sets EA_DUM on the Compustat report date; N = 6,683 / 8,608.

By the pre-registered standard, definitions explain part of the descriptive statistics but not Table 3: the distance falls clearly, yet the difference remains insignificant. The direction is notable, though. The lower-spread coefficient becomes insignificant while the higher-spread coefficient stays significant, and the paper's 1.501 now lies inside the confidence interval (p = 0.154). Table 3 is therefore sensitive to how a concurrent earnings forecast is defined.

## 8 Implications for the Paper's Conclusions

The paper has no exogenous variation in CAR. It relies instead on mechanism-based identification: it derives predictions that only the feedback mechanism would generate and tests them one by one. Table 3 is

the most important link, and the paper states explicitly that omitted variables or reverse causality would struggle to explain why the effect appears only for forecasts with rising spreads.

Three specifications give a consistent picture. The group difference is 0.21 in the main specification (p = 0.672), 1.08 from 2009 onward (p = 0.058) and 0.69 under the closest definitions (p = 0.222). It is always positive but smaller than in the paper, and never significant at the 5% level. The Table 3 evidence is therefore fragile in current data and depends on the sample period and on variable definitions. Two alternative explanations cannot be ruled out by Table 3: common information, such as oil prices that move both the stock price and capex plans, and market anticipation of the manager's later revision.

This assessment propagates along the chain of evidence (Appendix A). Table 5 E2a and E2b are the most exposed: the stronger feedback after mutual fund outflows is insignificant across all forecasts (p = 0.176 and 0.238) and significant only when spreads rise, and the paper's second hypothesis (Table 6) is motivated by this finding. Table 5 C is the most robust, being significant across all forecasts (p = 0.007). Table 4's numbers come from Table 2 and are unaffected, but labeling those adjustments "feedback-driven" still relies on Table 3.

A cautious statement is that market reactions to capex forecasts predict later capex adjustments, while current WRDS data provide no reliable evidence that this relation is confined to forecasts that attract informed trading.

## 9 Limitations

This report tests whether the paper replicates in current data; it is not a final verdict on the paper. First, forecast-level definitions can be inferred only from sample characteristics; the share of "others" and mean CAR remain unexplained, and resolving them requires the authors' sample list. Second, the early-year gap most likely arises at the paper's CRSP match, but the paper does not report that step by year. Third, our spreads come from CRSP daily closing quotes, whereas the paper does not state its source; the shares of rising spreads are similar (56.4% vs 55.2%), but equal shares need not mean the same forecasts in each group. Fourth, the partial patterns from 2009 onward and under the closest definitions are post hoc observations and serve only as leads. Fifth, Tables 4–6 have not been replicated, so Appendix A's assessment of them rests on the structure of the paper's own tables.

## Appendix A Evidence Map of the Paper

Figure A1 sorts every empirical test in the paper by whether it uses the Table 3 spread split. Tests on the left use the "higher spreads" subsample as the marker of feedback, so they inherit the Table 3 result directly. The regressions on the right do not use the split. This replication covers Tables 1–3 only; Tables 4–6 have not been replicated, so the classification below rests on the structure of the paper's own tables.




**Figure A1. Evidence map of the paper and the reach of Table 3**

**Table A1. Dependence of later tests on the Table 3 spread split**

| Test | Key comparison | All forecasts: p | Higher spreads: p | Dependence | Status |
|---|---|---|---|---|---|
| Table 3 cols 3–4 | Positive vs negative reactions | — | CAR_POS: 0.099 | Direct | Ours: 0.232 |
| Table 5 A | High vs low PIN | 0.076 | 0.058 | Direct (cols 1–2 marginal) | Not replicated |
| Table 5 B | Low vs high insider trading | 0.153 | 0.130 | Direct; insignificant in the paper | Not replicated |
| Table 5 C | Small vs large ERC (catering) | 0.007 | 0.020 | Holds in cols 1–2 | Not replicated |
| Table 5 D | Long vs short CEO horizon | 0.083 | 0.047 | Direct (2,969 forecasts) | Not replicated |
| Table 5 E2a | After large fund outflows vs other | 0.176 | 0.027 | Entirely through the split | Not replicated |
| Table 5 E2b | Same, outflow firms only | 0.238 | 0.032 | Entirely through the split | Not replicated |
| Table 4 | Feedback capex and performance | — | — | Numbers unaffected; fn. 27 direct | Not replicated |
| Table 5 E1 | Spreads around forecasts | — | — | None | Not replicated |
| Table 6 | Likelihood of forecasting (H2) | — | — | Regression unaffected; motivated by E2a | Not replicated |

Notes: All p-values are from the paper. "All forecasts" are columns (1)–(2) of each panel and "higher spreads" columns (5)–(6). A comparison that is significant only within higher spreads, and not across all forecasts, draws its evidence entirely from the Table 3 split.

This pins down how far the Table 3 result reaches. Table 5 E2a and E2b are the most exposed: the stronger feedback after mutual fund outflows is insignificant across all forecasts and significant only among forecasts with rising spreads, so the conclusion that managers rely more on disclosure-based feedback after noise-trading shocks rests entirely on the Table 3 split. The paper's second hypothesis (Table 6) is motivated by exactly this finding. Panel B is insignificant in the paper itself, and Panels A and D are only marginal across all forecasts. Panel C is the most robust: the ERC split is significant across all forecasts ($p = 0.007$) without the spread split. Table 4 uses the fitted values from Table 2 model (8), so its numbers are unaffected, but calling these adjustments "feedback-driven" still relies on Table 3.

# Appendix B Tables 1–3 in Full

**Table B1. Table 1 Panel A: descriptive statistics**

| Variable | Source | N | Mean | Median | SD | Min | Max |
|---|---|---|---|---|---|---|---|
| CAPEX_FORE | Paper | 17,577 | 586.539 | 120.000 | 1,318.429 | 2.000 | 8,000.000 |
| | Ours | 18,866 | 551.050 | 100.000 | 1,430.435 | 2.000 | 10,500.000 |
| CAPEX_ACT | Paper | 17,577 | 654.077 | 123.512 | 1,527.146 | 0.000 | 9,496.000 |
| | Ours | 18,866 | 574.861 | 96.703 | 1,505.327 | 0.517 | 10,861.000 |
| CAPEX_ADJ | Paper | 17,577 | 7.421 | 0.000 | 48.482 | −77.050 | 290.910 |
| | Ours | 18,866 | 3.945 | −0.326 | 44.512 | −88.676 | 261.366 |
| CAR | Paper | 17,577 | −0.148 | 0.069 | 8.974 | −31.359 | 25.350 |
| | Ours | 18,866 | 0.318 | 0.219 | 9.283 | −28.384 | 29.420 |
| SIZE | Paper | 17,577 | 7.378 | 7.344 | 1.629 | 1.366 | 11.300 |
| | Ours | 18,849 | 7.347 | 7.278 | 1.645 | 3.665 | 11.705 |
| LAG_CAPX | Paper | 17,546 | 611.045 | 118.307 | 1,398.011 | 0.000 | 8,620.081 |
| | Ours | 18,854 | 554.977 | 92.609 | 1,492.623 | 0.587 | 11,027.000 |
| SURP | Paper | 17,577 | −0.033 | 0.000 | 0.783 | −4.684 | 3.212 |
| | Ours | 18,840 | −0.016 | 0.000 | 0.815 | −4.258 | 3.436 |
| EA_DUM | Paper | 17,577 | 0.559 | 1.000 | 0.497 | 0.000 | 1.000 |
| | Ours | 18,866 | 0.662 | 1.000 | 0.473 | 0.000 | 1.000 |

Notes: All variables are winsorized at 1% and 99% in this replication (D14). LAG_CAPX is in $ millions.

**Table B2. Table 1 Panel B: forecasts and first-time forecasters by year**

| Year | Forecasts: paper | Forecasts: ours | Difference | First-time: paper | First-time: ours |
|---|---|---|---|---|---|
| 2003 | 22 | 42 | +20 | 22 | 40 |
| 2004 | 226 | 354 | +128 | 121 | 178 |
| 2005 | 488 | 639 | +151 | 181 | 222 |
| 2006 | 874 | 1,038 | +164 | 235 | 280 |
| 2007 | 841 | 972 | +131 | 197 | 242 |
| 2008 | 1,560 | 1,738 | +178 | 234 | 275 |
| 2009 | 2,064 | 2,231 | +167 | 225 | 290 |
| 2010 | 1,974 | 2,086 | +112 | 133 | 156 |
| 2011 | 2,048 | 2,139 | +91 | 122 | 144 |
| 2012 | 1,993 | 2,108 | +115 | 96 | 120 |
| 2013 | 1,931 | 1,959 | +28 | 87 | 87 |
| 2014 | 1,779 | 1,820 | +41 | 73 | 77 |
| 2015 | 1,777 | 1,740 | −37 | 64 | 69 |
| Total | 17,577 | 18,866 | +1,289 | 1,790 | 2,180 |

**Table B3. Table 1 Panel C: forecast types**

| Type | Paper: N | Paper: share | Ours: N | Ours: share |
|---|---|---|---|---|
| Point | 9,720 | 55.30% | 10,300 | 54.60% |
| Range | 6,061 | 34.48% | 8,044 | 42.64% |
| Others (open-ended) | 1,796 | 10.22% | 522 | 2.77% |
| Total | 17,577 | 100% | 18,866 | 100% |

**Table B4. Table 2 in full (dependent variable: CAPEX_ADJ)**

| Variable | Source | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|---|
| CAR | Paper | 1.193*** | 1.207*** | 1.196*** | 0.975*** | 0.975*** | 0.672* | 1.006*** | 0.849*** |
| | | (0.372) | (0.371) | (0.377) | (0.294) | (0.322) | (0.370) | (0.326) | (0.304) |
| | Ours | 1.193*** | 1.228*** | 1.276*** | 1.017*** | 1.017*** | 0.996* | 1.014*** | 0.988*** |
| | | (0.331) | (0.329) | (0.336) | (0.260) | (0.220) | (0.521) | (0.220) | (0.219) |
| SF_CAR | Paper | | | | | | 0.039 | | |
| | | | | | | | (0.061) | | |
| | Ours | | | | | | 0.393 | | |
| | | | | | | | (0.643) | | |
| MKTRET | Paper | | | | | | | −0.252 | −0.202 |
| | | | | | | | | (0.308) | (0.322) |
| | Ours | | | | | | | 0.161 | 0.169 |
| | | | | | | | | (0.322) | (0.322) |
| CAR_PRE | Paper | | | | | | | 0.305 | 0.230 |
| | | | | | | | | (0.200) | (0.197) |
| | Ours | | | | | | | 0.369 | 0.344 |
| | | | | | | | | (0.235) | (0.238) |
| SIZE | Paper | | 1.485** | 1.411** | 0.419 | 0.419 | 3.351 | 0.444 | −1.513 |
| | | | (0.632) | (0.653) | (1.302) | (1.128) | (2.203) | (1.131) | (1.415) |
| | Ours | | 0.743 | 0.708 | 2.080** | 2.080** | 4.074 | 2.093** | 1.139 |
| | | | (0.499) | (0.518) | (0.951) | (0.986) | (2.784) | (0.986) | (1.107) |
| LAG_CAPX | Paper | | | | | | | | −0.160*** |
| | | | | | | | | | (0.032) |
| | Ours | | | | | | | | −0.084*** |
| | | | | | | | | | (0.029) |
| SURP | Paper | | | | | | | | 0.213 |
| | | | | | | | | | (0.385) |
| | Ours | | | | | | | | 0.228 |
| | | | | | | | | | (0.343) |
| EA_DUM | Paper | | | | | | | | −0.108 |
| | | | | | | | | | (0.619) |
| | Ours | | | | | | | | −1.408 |
| | | | | | | | | | (1.104) |
| N | Paper | 17,577 | 17,577 | 17,577 | 17,577 | 17,577 | 3,456 | 17,577 | 17,505 |
| | Ours | 18,849 | 18,849 | 18,849 | 18,849 | 18,849 | 3,791 | 18,846 | 18,764 |
| Adj. $R^2$ | Paper | 0.001 | 0.003 | 0.007 | 0.523 | 0.523 | 0.648 | 0.523 | 0.526 |
| | Ours | 0.001 | 0.001 | 0.002 | 0.501 | 0.501 | 0.568 | 0.501 | 0.502 |
| Clustering | | Firm | Firm | Firm | Firm | Ind. | Ind. | Ind. | Ind. |
| Fixed effects | | None | None | YQ | Firm, YQ | Firm, YQ | Firm, YQ | Firm, YQ | Firm, YQ |

Notes: Clustered standard errors in parentheses. Our significance uses a t distribution with (clusters − 1) degrees of freedom; the paper's stars are as printed. CAR, SF_CAR, MKTRET and CAR_PRE are standardized. YQ = year-quarter. * p < 0.1, ** p < 0.05, *** p < 0.01.

**Table B5. Table 3 in full (dependent variable: CAPEX_ADJ)**

| Variable | Source | (1) Lower | (2) Higher | (3) Lower | (4) Higher |
|---|---|---|---|---|---|
| CAR | Paper | −0.295 | 1.206** | | |
| | | (0.372) | (0.497) | | |
| | Ours | 0.674** | 0.880** | | |
| | | (0.327) | (0.350) | | |
| CAR_POS | Paper | | | −0.203 | 0.893* |
| | | | | (0.419) | (0.455) |
| | Ours | | | 0.122 | 0.676* |
| | | | | (0.360) | (0.388) |
| CAR_NEG | Paper | | | −0.045 | 0.185 |
| | | | | (0.176) | (0.129) |
| | Ours | | | 0.803* | 0.412 |
| | | | | (0.455) | (0.394) |
| MKTRET | Paper | 0.103 | −0.327 | 0.104 | −0.323 |
| | | (0.539) | (0.654) | (0.538) | (0.655) |
| | Ours | −0.022 | 0.256 | −0.018 | 0.257 |
| | | (0.588) | (0.318) | (0.588) | (0.318) |
| CAR_PRE | Paper | 0.452 | 0.124 | 0.451 | 0.130 |
| | | (0.375) | (0.302) | (0.376) | (0.305) |
| | Ours | 0.061 | 0.349 | 0.060 | 0.349 |
| | | (0.301) | (0.347) | (0.302) | (0.346) |
| SIZE | Paper | n.r. | n.r. | n.r. | n.r. |
| | Ours | 0.407 | 1.382 | 0.318 | 1.412 |
| | | (1.839) | (1.074) | (1.834) | (1.091) |
| LAG_CAPX | Paper | n.r. | n.r. | n.r. | n.r. |
| | Ours | −0.117*** | −0.039 | −0.117*** | −0.039 |
| | | (0.042) | (0.055) | (0.042) | (0.056) |
| SURP | Paper | n.r. | n.r. | n.r. | n.r. |
| | Ours | 0.630 | 0.022 | 0.632 | 0.024 |
| | | (0.775) | (0.652) | (0.771) | (0.654) |
| EA_DUM | Paper | n.r. | n.r. | n.r. | n.r. |
| | Ours | −1.298 | −1.821 | −1.254 | −1.835 |
| | | (1.270) | (1.201) | (1.269) | (1.183) |
| N | Paper | 7,770 | 9,570 | 7,770 | 9,570 |
| | Ours | 8,135 | 10,542 | 8,135 | 10,542 |
| Adj. $R^2$ | Paper | 0.512 | 0.531 | 0.512 | 0.531 |
| | Ours | 0.503 | 0.515 | 0.503 | 0.515 |

Notes: n.r. = not reported; the paper does not report control coefficients in Table 3. Firm and year-quarter fixed effects; standard errors clustered by industry (67 clusters).

**Table B6. Table 3: p-values of the difference between lower and higher spreads**

| Variable | Columns | Paper | Ours |
|---|---|---|---|
| CAR | (1)–(2) | 0.019 | 0.672 |
| MKTRET | (1)–(2) | 0.656 | 0.588 |
| CAR_PRE | (1)–(2) | 0.540 | 0.556 |
| CAR_POS | (3)–(4) | 0.099 | 0.232 |
| CAR_NEG | (3)–(4) | 0.262 | 0.520 |
| MKTRET | (3)–(4) | 0.658 | 0.592 |
| CAR_PRE | (3)–(4) | 0.549 | 0.555 |

**Table B7. Robustness checks reported in the text (CAR coefficient, Table 2 model 8)**

| Specification | Paper | Ours | N |
|---|---|---|---|
| Baseline: Table 2 model (8) | 0.849*** | 0.988*** (0.219) | 18,764 |
| Ranges scaled at the midpoint (fn. 18) | robust | 0.987*** (0.235) | 18,764 |
| Point and range forecasts only (fn. 18) | robust | 0.979*** (0.206) | 18,243 |
| Control for asset growth (fn. 19) | robust | 0.960*** (0.219) | 18,745 |
| Industry × year fixed effects (fn. 20) | 0.800** | 0.779*** (0.241) | 18,764 |
| State × year fixed effects (fn. 20) | 0.968*** | 0.752*** (0.221) | 17,932 |
| Forecasts from 2009 on (fn. 24) | robust | 1.199*** (0.258) | 14,008 |

# Appendix C Annual Sample Waterfall

**Table C1. Annual sample waterfall (number of forecasts)**

| Year | S0 I/B/E/S raw | S1 one per announcement | S2 one per firm-quarter | S3 Compustat match | S4 CRSP reaction | S5 not with earnings forecast | Paper final |
|---|---|---|---|---|---|---|---|
| 2002 | 2 | 2 | 2 | 2 | 2 | 0 | — |
| 2003 | 173 | 152 | 151 | 146 | 146 | 42 | 22 |
| 2004 | 1,287 | 1,190 | 1,128 | 1,095 | 1,095 | 354 | 226 |
| 2005 | 2,207 | 2,038 | 1,896 | 1,798 | 1,797 | 639 | 488 |
| 2006 | 3,024 | 2,817 | 2,630 | 2,484 | 2,480 | 1,038 | 874 |
| 2007 | 2,731 | 2,477 | 2,273 | 2,112 | 2,110 | 972 | 841 |
| 2008 | 4,856 | 4,367 | 3,863 | 3,678 | 3,678 | 1,738 | 1,560 |
| 2009 | 5,344 | 4,899 | 4,132 | 3,933 | 3,932 | 2,231 | 2,064 |
| 2010 | 5,233 | 4,810 | 4,099 | 3,895 | 3,895 | 2,086 | 1,974 |
| 2011 | 5,407 | 4,921 | 4,169 | 3,948 | 3,948 | 2,139 | 2,048 |
| 2012 | 5,320 | 4,866 | 4,157 | 3,976 | 3,976 | 2,108 | 1,993 |
| 2013 | 5,278 | 4,775 | 4,044 | 3,855 | 3,855 | 1,959 | 1,931 |
| 2014 | 5,015 | 4,555 | 3,872 | 3,725 | 3,724 | 1,820 | 1,779 |
| 2015 | 4,760 | 4,342 | 3,654 | 3,548 | 3,548 | 1,740 | 1,777 |
| 2016 | 4,128 | 3,788 | 3,352 | 3,257 | 3,257 | 1,478 | — |
| Total | 54,765 | 49,999 | 43,422 | 41,452 | 41,443 | 20,344 | 17,577 |

Notes: By announcement year. Keeping 2003–2015 after S5 gives the final sample. The paper reports only its final counts by year (Table 1 Panel B) and the all-year totals at S3–S5 (40,785; 36,900; 19,427).

**Table C2. Retention rates R(t, k) = N(t, k) / N(t, k−1) and ratios to the paper**

| Year | S1 | S2 | S3 | S4 | S5 | S5 (EPS only) | Paper ÷ S4 | Paper ÷ S5 |
|---|---|---|---|---|---|---|---|---|
| 2003 | 0.879 | 0.993 | 0.967 | 1.000 | 0.288 | 0.418 | 0.151 | 0.524 |
| 2004 | 0.925 | 0.948 | 0.971 | 1.000 | 0.323 | 0.487 | 0.206 | 0.638 |
| 2005 | 0.923 | 0.930 | 0.948 | 0.999 | 0.356 | 0.503 | 0.272 | 0.764 |
| 2006 | 0.932 | 0.934 | 0.944 | 0.998 | 0.419 | 0.501 | 0.352 | 0.842 |
| 2007 | 0.907 | 0.918 | 0.929 | 0.999 | 0.461 | 0.570 | 0.399 | 0.865 |
| 2008 | 0.899 | 0.885 | 0.952 | 1.000 | 0.473 | 0.584 | 0.424 | 0.898 |
| 2009 | 0.917 | 0.843 | 0.952 | 1.000 | 0.567 | 0.653 | 0.525 | 0.925 |
| 2010 | 0.919 | 0.852 | 0.950 | 1.000 | 0.536 | 0.636 | 0.507 | 0.946 |
| 2011 | 0.910 | 0.847 | 0.947 | 1.000 | 0.542 | 0.642 | 0.519 | 0.957 |
| 2012 | 0.915 | 0.854 | 0.956 | 1.000 | 0.530 | 0.627 | 0.501 | 0.945 |
| 2013 | 0.905 | 0.847 | 0.953 | 1.000 | 0.508 | 0.614 | 0.501 | 0.986 |
| 2014 | 0.908 | 0.850 | 0.962 | 1.000 | 0.489 | 0.603 | 0.478 | 0.977 |
| 2015 | 0.912 | 0.842 | 0.971 | 1.000 | 0.490 | 0.617 | 0.501 | 1.021 |

Notes: t is the year, k the filtering step and N the number of forecasts retained at that step. The last two columns divide the paper's final count by our counts at S4 and S5. Share of 2003–2009 at each step: raw records 0.388, S4 0.399, S5 0.372, paper final 0.346.

# Appendix D All Variants of Forecast-Level Definitions

**Table D1. Forecast-level definitions: all variants**

| Specification | N | Firms | Dropped | EA_DUM | Range | Others | CAPEX_ADJ mean | CAPEX_ADJ Median | CAR mean | D |
|---|---|---|---|---|---|---|---|---|---|---|
| Paper | 17,577 | 1,790 | 47.4% | 0.559 | 34.5% | 10.2% | 7.421 | 0.000 | −0.148 | — |
| Current specification | 18,866 | 2,180 | 50.9% | 0.662 | 42.6% | 2.8% | 3.945 | −0.326 | 0.318 | 1.484 |
| Type: codes 06, 08 as others | 18,866 | 2,180 | 50.9% | 0.662 | 42.4% | 3.0% | 3.964 | −0.326 | 0.318 | 1.477 |
| Type: by I/B/E/S code only | 18,866 | 2,180 | 50.9% | 0.662 | 42.4% | 3.0% | 3.964 | −0.326 | 0.318 | 1.477 |
| EA_DUM: I/B/E/S date, same day | 18,866 | 2,180 | 50.9% | 0.665 | 42.6% | 2.8% | 3.945 | −0.326 | 0.318 | 1.487 |
| EA_DUM: RDQ ±1 day | 18,866 | 2,180 | 50.9% | 0.899 | 42.6% | 2.8% | 3.945 | −0.326 | 0.318 | 1.721 |
| Earnings forecast: EPS only | 22,961 | 2,456 | 39.9% | 0.682 | 44.8% | 2.6% | 2.867 | −0.385 | 0.287 | 2.031 |
| Earnings forecast: any other guidance | 15,387 | 1,908 | 59.9% | 0.633 | 40.5% | 2.9% | 5.557 | 0.000 | 0.290 | 1.087 |
| Earnings forecast: earnings-type ±1 day | 16,917 | 1,982 | 56.0% | 0.732 | 42.3% | 2.9% | 4.346 | −0.352 | 0.287 | 1.318 |
| Closest (any other guidance + by code) | 15,387 | 1,908 | 59.9% | 0.633 | 40.3% | 3.1% | 5.580 | 0.000 | 0.290 | 1.080 |

Notes: "Dropped" is the share removed at the step that deletes forecasts issued on the same day as an earnings forecast. The full grid has 48 combinations (4 × 3 × 4); see `output/definitions_grid.csv`.

# Appendix E Full Results of the Diagnostic Tests

**Table E1. Split tests (CAR coefficient, lower vs higher spreads)**

| Test | N | Lower | Higher | Difference (SE) | p | 95% CI |
|---|---|---|---|---|---|---|
| A Main specification | 8,135 / 10,542 | 0.674** | 0.880** | 0.206 (0.485) | 0.672 | [−0.76, 1.18] |
| A2 CAR_POS | 8,135 / 10,542 | 0.122 | 0.676* | 0.553 (0.458) | 0.232 | [−0.36, 1.47] |
| C Top vs bottom tercile | 6,213 / 6,224 | 0.484 | 1.043* | 0.560 (0.714) | 0.436 | [−0.87, 1.98] |
| D1 2009 onward | 6,029 / 7,978 | 0.341 | 1.422*** | 1.081 (0.560) | 0.058 | [−0.04, 2.20] |
| D2 Point and range only | 7,904 / 10,252 | 0.525 | 0.986*** | 0.460 (0.501) | 0.361 | [−0.54, 1.46] |
| D3 Midpoint scaling | 8,135 / 10,542 | 0.590* | 0.903** | 0.314 (0.497) | 0.531 | [−0.68, 1.31] |
| E Common shares only | 7,265 / 9,390 | 0.913** | 0.698* | −0.215 (0.541) | 0.692 | [−1.30, 0.86] |

Notes: Differences are estimated in a fully interacted pooled regression with a joint clustered covariance; 67 industry clusters, t distribution with 66 degrees of freedom. Paper benchmarks: 1.501 for CAR and 1.096 for CAR_POS.

**Table E2. Continuous test (B)**

| Variable | Coefficient | SE | p |
|---|---|---|---|
| CAR | 0.951 | 0.218 | < 0.001 |
| CAR × ΔSpread | 0.071 | 0.264 | 0.789 |
| ΔSpread | −0.393 | 0.175 | 0.028 |

Notes: ΔSpread is the log ratio of event-window to pre-event relative spreads, winsorized and standardized. N = 18,677. Controls and fixed effects as in Table 2 model (8).

**Table E3. Characteristics of sample variants**

| Sample | Forecasts | Firms | EA_DUM | Range share | CAR mean | D |
|---|---|---|---|---|---|---|
| Paper | 17,577 | 1,790 | 0.559 | 0.345 | −0.148 | — |
| Our main sample | 18,866 | 2,180 | 0.662 | 0.426 | 0.318 | 1.484 |
| Common shares (shrcd 10/11) | 16,819 | 1,855 | 0.662 | 0.436 | 0.344 | 1.354 |
| Exchanges 1–3 | 18,851 | 2,180 | 0.662 | 0.426 | 0.322 | 1.490 |
| CRSP history ≥ 12 months | 18,618 | 2,154 | 0.662 | 0.426 | 0.313 | 1.453 |
| I/B/E/S link score ≤ 1 | 18,825 | 2,169 | 0.662 | 0.427 | 0.326 | 1.483 |
| Common + exchange + history | 16,612 | 1,836 | 0.662 | 0.436 | 0.343 | 1.359 |
| Point in time (activated ≤ 2018-07-21) | 18,866 | 2,180 | 0.662 | 0.426 | 0.318 | 1.484 |

Notes: The distance D was defined before any run and combines N, firms, EA_DUM, forecast-type shares and the means of CAPEX_ADJ and CAR. The variant with the smallest D (common shares) is used in test E.

# Appendix F Implementation Choices Not Stated in the Paper

**Table F1. Implementation choices relevant to Tables 1–3**

| No. | Issue | This replication | Evidence or alternative |
|---|---|---|---|
| D1 | I/B/E/S measure codes | CPX (capex), SAL (sales), EPS | Confirmed from the codes in the data |
| D2 | Start of the sample | Keep 2003–2015 | Only 2 raw capex forecasts in 2002; the paper's Table 1 Panel B starts in 2003 |
| D3 | Open-ended forecasts | range_desc 03, 04, 10–13, 15–17 open-ended; 01, 06, 08 range; 02, 14 point | I/B/E/S Guidance User Guide (2009), Appendix 2 |
| D4 | Units | millions ×1, billions ×1,000 | Median actual-to-forecast ratio 0.99 |
| D5 | Several fiscal years in one announcement | Keep the nearest year; drop horizons over 24 months | Not stated in the paper |
| D6 | Several forecasts in a quarter | Keep the first; collapse by (I/B/E/S ticker, calendar quarter) before filtering | The paper's waterfall arithmetic is exact, so no later de-duplication |
| D7 | Denominator for range forecasts | Midpoint of the range | Alternative: the violated bound |
| D8 | CAPEX_ADJ for open-ended forecasts | Treated as a point forecast at val_1 | Alternative: one-sided |
| D9 | "Concurrent with an earnings forecast" | Any earnings-type guidance on the same day (EPS, GAAP EPS, net income, pretax, EBITDA, etc.) | Closest final N among four definitions; EPS only as a robustness check |
| D10 | Timing of EA_DUM | Same day as the Compustat report date (RDQ) | Same day 0.662, ±1 day 0.899, paper 0.559 |
| D11 | After-hours releases | Day 0 not shifted | Covered by the 5-day window |
| D12 | Missing returns in CAR | All 5 daily returns required | — |
| D13 | SF_CAR and CAR on the same day | Windows at least 5 trading days apart | Matches the paper's 3,456 sales forecasts and 86-day median gap |
| D14 | Winsorization | All Table 1 variables at 1% / 99% | Paper extremes match our percentiles (e.g. CAPEX_ACT max 9,496) |
| D15 | Units of LAG_CAPX | Percent of market value in regressions | The −0.160 coefficient is plausible only in percent |
| D17 | Spread units | Percent | Matches the 0.073 median non-event spread |
| D18 | "Industry" for clustering | Two-digit SIC, modal per firm | Firm fixed effects must nest within clusters |
| D23 | Share-code filter | None | Not stated; common shares used as a diagnostic variant |
| D25 | Singleton observations | Kept | Table 2 models (1)–(5) have identical N |
| D26 | Degrees-of-freedom correction | reghdfe convention | Alternative: areg |
| D27 | Sample for standardization | Full sample once; not redone within splits | Redoing within splits changes coefficient scale |
| D32 | SURP across stock splits | Lagged EPS restated to the current share basis with ajexq | Not stated; avoids spurious surprises in split quarters |
| D35 | CAR_POS / CAR_NEG | Split at zero, then standardize each | Alternative: standardize, then split |

Notes: The other 14 choices (D16, D19–D22, D24, D28–D31, D33, D34, D36, D37) concern Tables 4–6 only and are not used here. Each choice is a switch in `config.py`.

# Appendix G Code and Run Order

**Table G1. Main files**

| File | Purpose |
|---|---|
| `config.py` | Switches for the 37 implementation choices (D1–D37) |
| `build_capex_sample.py`, `tables.py` | Sample construction and Tables 1–3 |
| `t3_sensitivity.py` | Table 3 under alternative choices |
| `diagnose_gap.py` | Sample-gap diagnostics (sample characteristics only) |
| `t3_diagnostics.py` | Pre-registered Table 3 diagnostics |
| `annual_waterfall.py` | Annual sample waterfall and retention rates |
| `diagnose_definitions.py` | Forecast-level definitions |
| `docs/` | Two pre-registrations: Table 3 diagnostics; forecast-level definitions |
| `tests/` | Estimator tests and end-to-end tests on synthetic data |

Notes: Code at github.com/ZQ-77777/jw2020-capex-feedback-replication. The repository contains no WRDS raw data.

Run order:

```
python run_all.py
python t3_sensitivity.py
python diagnose_gap.py
python t3_diagnostics.py
python annual_waterfall.py
python diagnose_definitions.py --pull
```